FCT SERVICE NOTE
SN06102022A

Class A Service information enclosed.

Date:

Subject: VCT 48v Fuse Board DC Power Screws Shorting to Chassis

Effectivity: Lightspeed 7.X (VCT) and PET Discovery VCT

Details: A short circuit could occur if the DC cable bolts are installed incorrectly.

Resolution:
See Illustrations 1, 2 and 3 for bolt clearances and further explanation.

Illustration 1: Rotating 48V Power Supply (Full Side View)

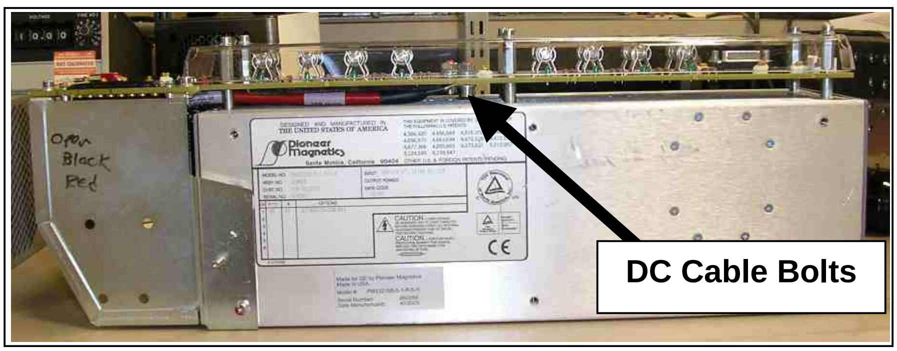


Illustration 2: Correctly Installed Bolts (Close Up View)

√ **Bolts Installed correctly** | X **Bolts installed Incorrectly**

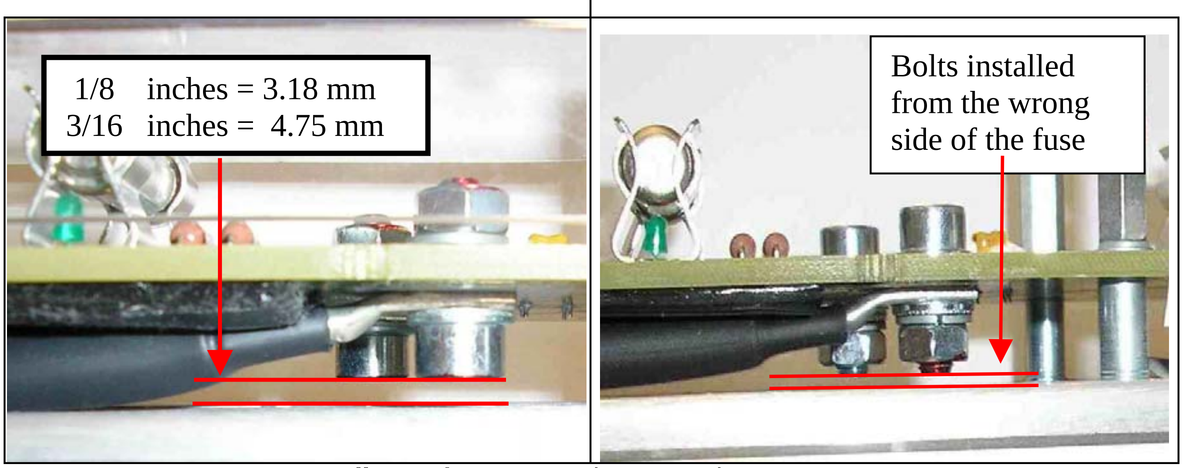


Illustration 3: Gap clearance close up.

Cause: The bolt orientation is critical to avoid a possible short circuit between the bolt and the power supply chassis. A possible Short Circuit condition is caused when the bolt is installed from the wrong side of the fuse board. A combination of PC board thickness, bolt length and washer thickness can all attribute to how far the screw extends past the pc board. See bolt to chassis clearance differences between Illustration 2 versus 3.

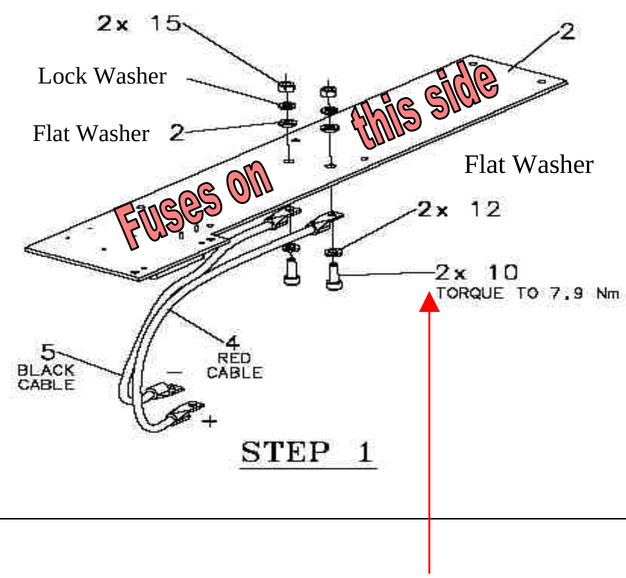
Based on field reports we have seen the 48v power supplies mounting screws short to the chassis immediately after installing FMI 25382. The root cause was improperly installed Bolts (Bolts used to secure DC power cables to the fuse boards). Several FMI kits were found to have the bolts assembled improperly, See Illustration 4 for proper assembly of the bolt, washer, cable and nut.

Illustration 4: Correct Mounting Hardware – Engineering drawing

1) Bolt orientation direction is critical

2) Lock Washer & flat Washer location are also critical

3) If bolts were removed Torque to 7.9 NM (70 in-lb) (as shown in Illustration 4).



Resolution: If FMI 25382 has not been completed. Inspect and correct the Fuse Board connection before installing FMI 25382. IF FMI 25382 was completed, inspect 48v power bolt orientation on the next visit. See Illustration 4 for proper assembly of the bolt, washer, cable and nut. Follow the drawing in Illustration 4. If bolts need to be removed, remember to torque them to 7.9 Nm (70 in-lb).

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